

First-Time Setup for OPTI-GAS

This guide is for someone who pulled or cloned the project for the first time.

1. Clone the repository

```
git clone https://github.com/drei-cabal/Opti-Gas.git
cd Opti-Gas
```

2. Install Python

Use Python `3.12` or newer.

Official download:

<https://www.python.org/downloads/>

During installation on Windows:

- enable `Add python.exe to PATH`
- make sure `pip` is included

Verify installation:

```
python --version
```

If `python` does not work, try:

```
py --version
```

3. Create and activate a virtual environment

Using `python`:

```
python -m venv .venv
.venv\Scripts\Activate.ps1
```

Or using `py`:

```
py -m venv .venv
.venv\Scripts\Activate.ps1
```

If PowerShell blocks activation:

```
Set-ExecutionPolicy -Scope Process Bypass
```

```
.venv\Scripts\Activate.ps1
```

4. Install project dependencies

```
pip install -r requirements.txt
```

5. Create the environment file

Copy the sample file:

```
copy .env.example .env
```

6. Get an OpenRouteService API key

This project uses OpenRouteService for route distance and estimated travel time.

OpenRouteService website:

<https://openrouteservice.org/>

General process:

1. Create an account
2. Sign in
3. Go to the dashboard/API keys section
4. Create a key for directions/routing use
5. Copy the generated API key

7. Add the API key to `.env`

Open `.env` and set:

```
ORS_API_KEY=your_real_api_key_here
```

You may also keep or adjust these if needed:

```
FLASK_HOST=0.0.0.0  
FLASK_PORT=5000  
FLASK_DEBUG=0
```

Important:

- never commit `.env`

- ``.env.example`` is safe to commit
- if you accidentally push a real key, rotate it immediately

8. Run the app

```
python app.py
```

Then open:

```
http://127.0.0.1:5000
```

9. Validate station data

Before editing or testing station data, run:

```
python scripts\validate_stations.py
```

Expected success output:

```
OK: 76 stations loaded
OK: All required fields present
OK: All coordinates within Tagum City bounds
OK: Duplicate station names are allowed
```

10. If Python is not available locally

This workspace has also been run using a bundled Python path on this machine:

```
C:\Users\ANDREI\.cache\codex-runtimes\codex-primary-runtime\dependencies\python\python.exe
```

Run with:

```
$env:PYTHONPATH='.vendor'
& 'C:\Users\ANDREI\.cache\codex-runtimes\codex-primary-runtime\dependencies\python\python.exe' app.py
```

This is machine-specific and should not be treated as the standard setup for other users.

11. Common first-time issues

``python` not found`

Use:

```
py app.py
```

or reinstall Python with PATH enabled.

API key missing

If `ORS\_API\_KEY` is empty, the app can still run, but routing may fall back to OSRM or local estimates instead of ORS.

Slow first load

The first load may be slower because of:

- browser geolocation
- first route calculations
- network latency to routing services

Later refreshes should be faster because of browser-side session reuse and backend route caching.

Port already in use

Run on another port:

```
$env:FLASK_PORT="5001"
python app.py
```

Then open:

```
http://127.0.0.1:5001
```

12. Recommended first-time command sequence

```
git clone https://github.com/drei-cabal/Opti-Gas.git
cd Opti-Gas
python -m venv .env
.venv\Scripts\Activate.ps1
pip install -r requirements.txt
copy .env.example .env
python app.py
```